

TOKUSHIMA AB, Japan

WMO: 478810

Lat: 34.133N

Lon: 134.607E

Elev: 8

StdP: 101.23

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.1	(c) 1.1	(d) -8.1	(e) 1.9	(f) 3.4	(g) -6.9	(h) 2.1	(i) 4.2	(j) 12.3	(k) 6.3	(l) 11.3	(m) 5.9	(n) 4.6	(o) 290	(p) 0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.9	(c) 33.0	(d) 25.6	(e) 31.2	(f) 25.6	(g) 30.2	(h) 25.4	(i) 27.1	(j) 30.0	(k) 26.7	(l) 29.6	(m) 26.3	(n) 29.1	(o) 4.5	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	21.7	28.7	26.0	21.4	28.5	25.2	20.4	28.1	85.8	30.3	83.9	30.0	82.1	29.0	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.6	(b) 10.1	(c) 9.0	(e) -1.4	(f) 35.9	(g) 1.0	(h) 1.2	(i) -2.1	(j) 36.8	(k) -2.7	(l) 37.5	(m) -3.2	(n) 38.2	(o) -3.9	(p) 39.1		
(a) 11.6	(b) 10.1	(c) 9.0	(e) -3.5	(f) 28.0	(g) 1.2	(h) 0.7	(i) -4.4	(j) 28.5	(k) -5.1	(l) 28.9	(m) -5.8	(n) 29.4	(o) -6.7	(p) 29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	6.1	6.7	9.7	14.7	19.3	22.8	26.6	28.0	24.7	19.4	13.7	8.6	(6)
(7)		DBStd	7.91	2.10	2.53	2.80	2.86	2.10	1.91	2.04	1.64	2.39	2.66	2.96	2.66	(7)
(8)		HDD10.0	327	122	98	41	2	0	0	0	0	0	0	3	60	(8)
(9)		HDD18.3	1562	379	327	268	113	13	0	0	0	0	18	141	303	(9)
(10)		CDD10.0	2787	1	5	32	143	290	383	514	557	441	291	115	15	(10)
(11)		CDD18.3	981	0	0	0	3	45	133	256	298	191	52	2	0	(11)
(12)		CDH23.3	7846	0	0	0	8	116	521	2298	3300	1448	156	0	0	(12)
(13)		CDH26.7	2241	0	0	0	0	11	64	653	1177	326	9	0	0	(13)
(14)	Wind	WSAvg	4.3	4.7	4.6	4.6	4.5	4.3	3.8	3.9	4.2	4.2	4.1	4.2	4.6	(14)
(15)	Precipitation	PrecAvg	1657	47	65	98	109	151	204	175	188	271	173	105	68	(15)
(16)		PrecMax	2495	113	147	179	226	433	382	404	952	849	469	282	249	(16)
(17)		PrecMin	992	7	14	46	41	45	55	63	22	68	19	15	6	(17)
(18)		PrecStd	410	29	36	37	49	94	76	84	196	176	111	70	52	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	17.0	19.8	24.8	28.0	30.1	35.0	35.2	32.1	28.0	22.2	18.1	(19)
(20)			MCWB	10.5	13.9	13.7	16.7	17.8	23.7	26.4	26.1	24.6	23.1	17.6	14.8	(20)
(21)		2%	DB	12.8	14.2	17.8	22.0	26.0	28.2	32.9	33.8	30.8	26.1	21.1	16.0	(21)
(22)			MCWB	9.3	10.9	13.3	15.5	17.4	22.2	25.7	25.9	24.7	21.7	17.3	12.6	(22)
(23)		5%	DB	11.2	12.8	16.2	20.2	24.2	27.0	31.0	32.1	29.8	25.0	20.0	14.2	(23)
(24)			MCWB	7.4	9.5	12.1	15.2	18.0	21.9	25.6	25.6	24.2	20.6	16.3	10.5	(24)
(25)		10%	DB	10.1	11.1	15.0	19.2	23.1	26.0	29.9	31.0	28.3	24.0	18.9	13.1	(25)
(26)			MCWB	6.5	7.6	11.3	14.6	17.7	21.8	25.5	25.9	23.7	19.7	15.5	9.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	14.7	16.2	19.4	22.2	25.6	27.5	28.0	27.1	24.7	19.7	15.7	(27)
(28)			MADB	13.6	16.5	17.8	21.4	24.0	28.1	30.5	30.5	29.3	26.5	21.0	17.1	(28)
(29)		2%	WB	9.9	11.8	14.6	17.7	20.8	24.6	26.9	27.3	26.4	23.1	18.3	13.2	(29)
(30)			MADB	11.4	13.9	16.5	19.7	22.8	26.6	29.8	30.2	28.6	25.2	20.3	15.6	(30)
(31)		5%	WB	8.2	10.1	13.2	16.7	19.9	23.9	26.4	26.9	25.7	21.7	17.1	11.5	(31)
(32)			MADB	10.5	11.7	15.4	19.2	22.2	25.7	29.3	30.0	28.0	24.0	19.3	13.7	(32)
(33)		10%	WB	6.8	8.3	11.7	15.7	19.1	23.2	26.0	26.6	25.0	20.3	15.7	10.1	(33)
(34)			MADB	9.4	10.7	14.6	18.3	21.6	24.9	28.7	29.5	27.5	22.9	18.3	12.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.5	6.8	7.6	7.7	7.1	5.6	5.3	5.9	6.0	6.6	6.9	6.7	(35)
(36)			MADB	8.1	8.9	9.3	9.0	9.1	7.3	7.3	7.5	7.1	7.1	7.6	8.2	(36)
(37)			MCWBR	6.0	6.8	6.8	5.4	4.7	3.4	3.0	3.1	3.5	4.3	5.2	6.0	(37)
(38)		5% WB	MADB	7.6	8.0	8.4	7.1	6.3	5.1	6.0	6.0	5.3	5.9	6.9	7.7	(38)
(39)			MCWBR	6.6	7.3	7.5	5.5	4.6	3.1	3.0	3.2	3.3	4.5	5.6	6.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.398	0.446	0.506	0.539	0.544	0.561	0.561	0.532	0.495	0.456	0.422	0.393	(40)	
(41)		taud	2.177	2.060	1.914	1.887	1.943	1.977	2.026	2.096	2.170	2.195	2.196	2.209	(41)	
(42)		Ebn at Noon	783	784	767	764	765	750	747	762	771	767	752	763	(42)	
(43)		Edh at Noon	117	146	183	197	188	182	172	158	141	127	114	108	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.55	3.14	4.10	4.87	5.21	4.63	5.12	5.30	4.11	3.35	2.61	2.29	(44)	
(45)		RadStd	0.18	0.24	0.29	0.47	0.51	0.47	0.63	0.48	0.36	0.34	0.22	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47) Regional (31 neighbors)		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A

Nomenclature: See separate page